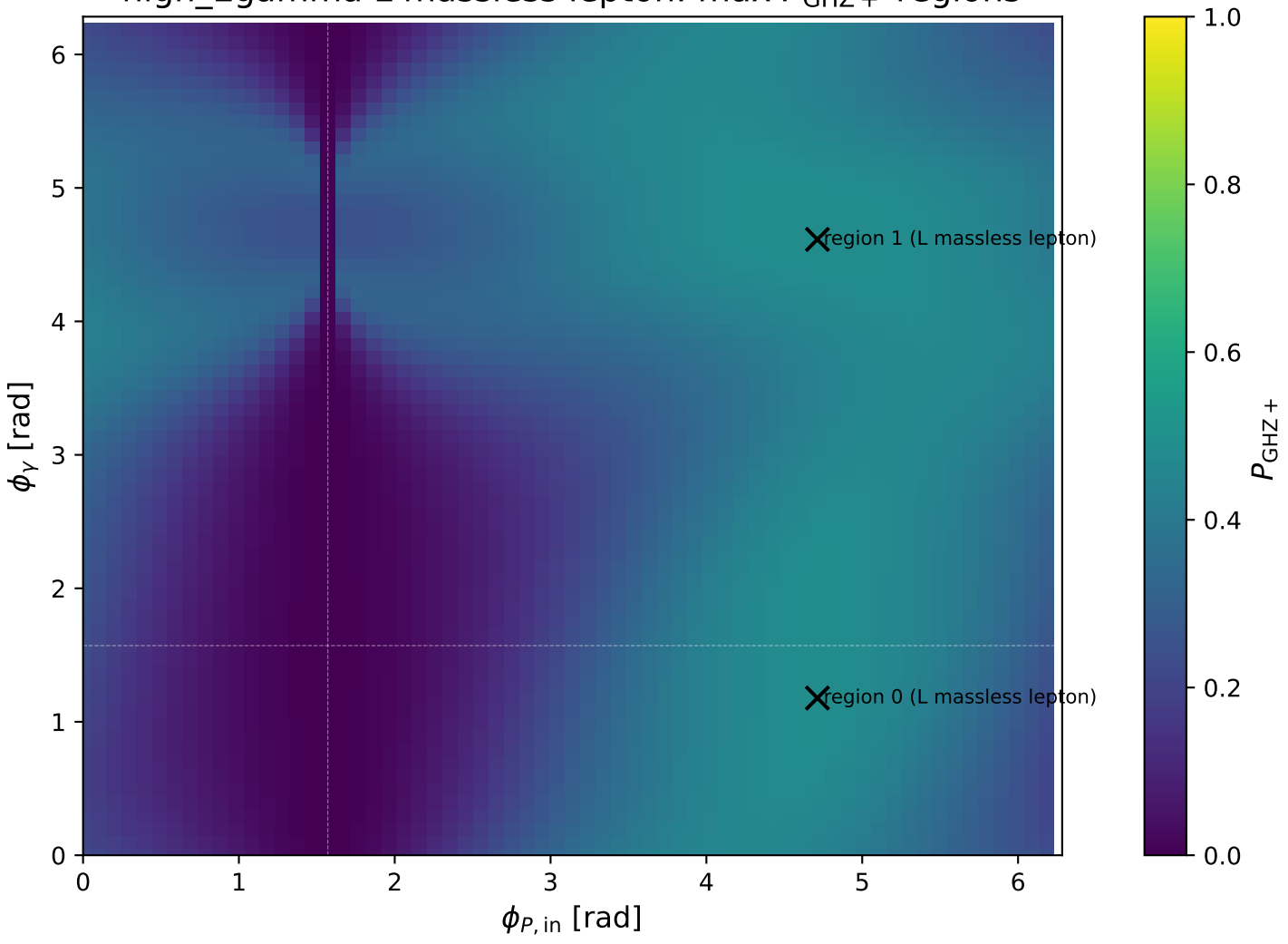
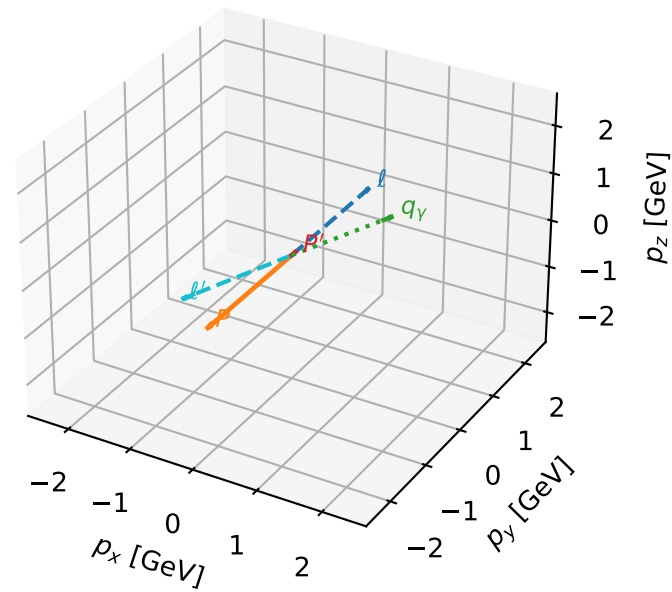


high_Egamma L massless lepton: max P_{GHZ} + regions



L massless lepton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

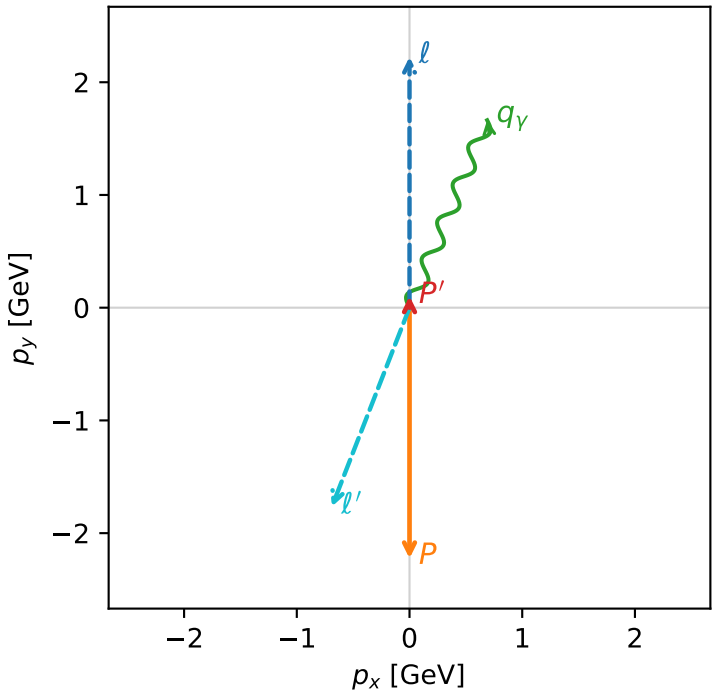


ghz_purity_high_egamma_l_lepton_region_0 $P_{\text{GHZ+}}=0.482713$
region: high_Egamma L_lepton_region_0
outgoing-state purity: 0.997848
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2} \sum_{h_p} \rho(L_{\ell_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.102337$
 $\theta_{\text{in}}=1.57079$, $\phi_{\ell}=1.5708$, $\phi_P=4.71239$
 $E_Y=1.8$, $\phi_Y=1.1781$
 $Q^2=16.2839$, $x_B=0.841965$, $t=-3.25132$, $W^2=3.93631$, $y=0.937217$
 $\theta(\ell', \gamma)=178.816$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.8950$ $p_x= -0.6888$ $p_y= -1.7653$ $p_z= 0.0000$
 P' $E= 0.9436$ $p_x= 0.0000$ $p_y= 0.1023$ $p_z= 0.0000$
 q_Y $E= 1.8000$ $p_x= 0.6888$ $p_y= 1.6630$ $p_z= 0.0000$

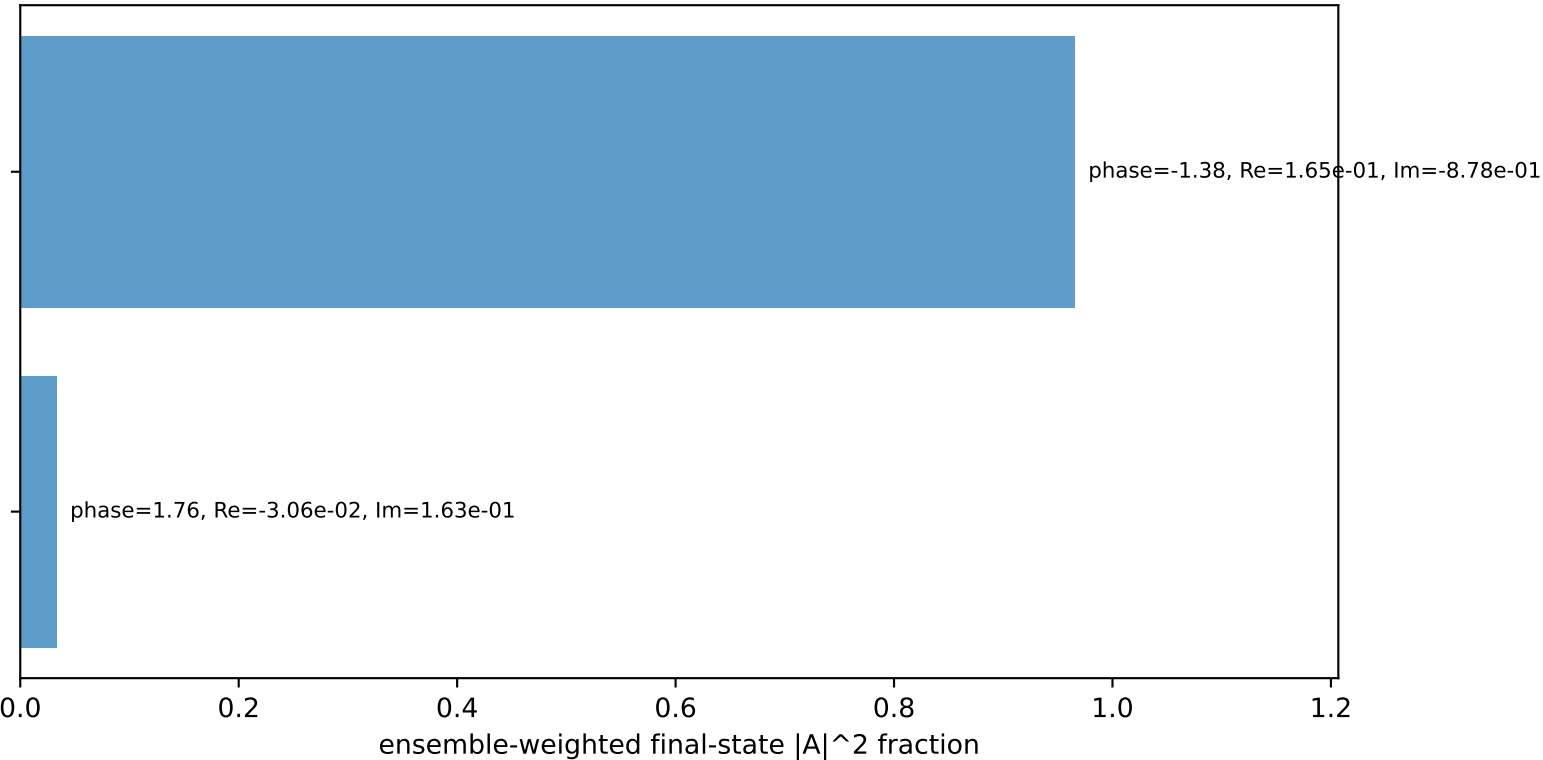
Transverse plane



$h_e=+1, h_p=+1, h_Y=+1$
electron L, proton $h=+1$

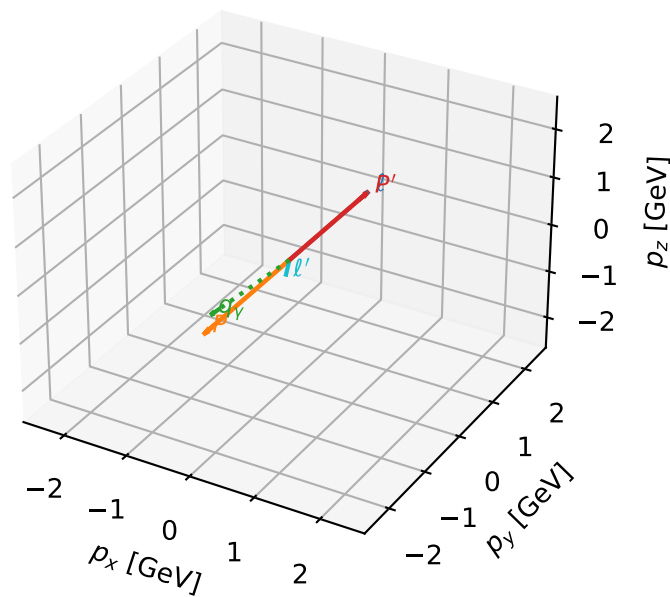
$h_e=+1, h_p=+1, h_Y=-1$
electron L, proton $h=+1$

Leading initial-ensemble/final-state components (N=2, retained=99.9%)



L massless lepton characteristic max P_{GHZ} + configuration

3D momenta

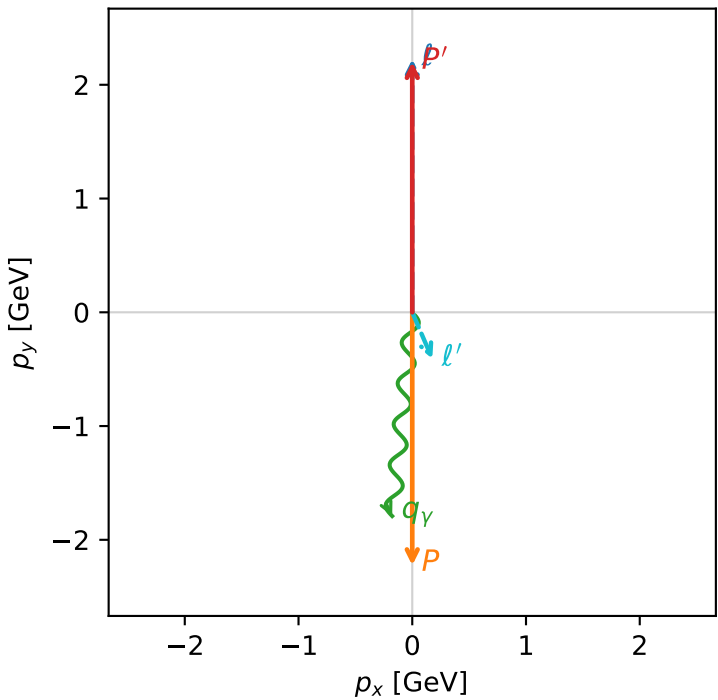


ghz_purity_high_egamma_l_lepton_region_1 $P_{\{\rm GHZ\}}=0.482463$
region: high_Egamma_L_lepton_region_1
outgoing-state purity: 0.999865
kinematic point: high_s_high_theta_in_high_Egamma
incoming state: $\frac{1}{2}\sum_{h_p}\rho(L_{l_0}, h_p)$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=2.20096$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1.8$, $\phi_\gamma=4.61421$
 $Q^2=3.80661$, $x_B=0.187472$, $t=-19.5835$, $W^2=17.3782$, $y=0.983958$
 $\theta(l', \gamma)=28.9268$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 l' $E= 0.4460$ $p_x= 0.1764$ $p_y= -0.4096$ $p_z= 0.0000$
 P' $E= 2.3925$ $p_x= 0.0000$ $p_y= 2.2010$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.1764$ $p_y= -1.7913$ $p_z= 0.0000$

Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
electron L, proton $h=+1$

$h_e=+1, h_p=+1, h_\gamma=-1$
electron L, proton $h=+1$

Leading initial-ensemble/final-state components (N=2, retained=100.0%)

